

JOSHUA NWACHINEMERE

AI Engineer | Python, Retrieval Systems, Multimodal AI & ML Evaluation

joshua0nwachinemere@gmail.com | +234 913 148 0096

github.com/dk3yyyy | [linkedin.com/in/joshua-nwachinemere](https://www.linkedin.com/in/joshua-nwachinemere) | joshua-nwachinemere.pages.dev

PROFILE

AI Engineer building Python systems for retrieval, multimodal inference, speech, model APIs, and ML evaluation. Built local review intelligence with citation validation and benchmarked evaluation, a privacy-oriented macOS assistant, and a temporal forecasting pipeline. Contributed eight merged fixes and tests across AI and data infrastructure projects, including OpenAI Agents SDK and Pydantic AI Harness.

SELECTED AI & ML PROJECTS

Local Review Intelligence | Local Retrieval & Evaluation

Python, Ollama, ChromaDB, LangChain, Streamlit

- Built a local-first review intelligence application with adaptive CSV mapping, content-addressed Chroma storage, reconciled indexing, a Streamlit dashboard, and an installable CLI.
- Implemented stable source IDs, validated citations, bounded answer repair, safe abstention, and a versioned 30-case evaluation comparing semantic retrieval with BM25.
- Source: https://github.com/dk3yyyy/local_AI_agent

Volyx Lens | Privacy-First Context-Aware AI Assistant

JavaScript, Electron, Swift, Azure AI Foundry, Multimodal APIs

- Built and hardened a privacy-oriented macOS assistant combining selected screen context, microphone input, meeting audio, local OCR, relevance-ranked retrieval, and multiple AI providers.
- Implemented bounded context memory, provider routing, explicit data-sharing controls, sandboxed Electron boundaries, automated tests, secret scanning, and release checks.
- Source: <https://github.com/dk3yyyy/volyx-lens>

Football Forecasting Lab | Temporal ML Evaluation Pipeline

Python, XGBoost, scikit-learn, FastAPI, Streamlit, SQLAlchemy

- Built data scrapers, temporal feature engineering, XGBoost outcome and Poisson goal models, a FastAPI service, and a Streamlit dashboard.
- Evaluated with rolling-origin testing across 1,140 Premier League matches: 53.77% outcome accuracy versus a 56.70% bookmaker-implied benchmark; the model did not outperform the benchmark.
- Source: https://github.com/dk3yyyy/football_predictor

OPEN-SOURCE CONTRIBUTIONS

OpenAI Agents Python SDK | Merged PR #3991

Python, WebSockets, retry policies, pytest | Jul 2026

- Made transient pre-response WebSocket server errors follow the SDK's retry policy while preserving non-transient handling and replay-safety checks. PR: <https://github.com/openai/openai-agents-python/pull/3991>

Pydantic AI Harness | Merged PR #503

Python, agent recovery, Code Mode, pytest | Jul 2026

- Surfaced unexpected Code Mode execution failures to the model as retries after resetting invalid session state, enabling recovery from lost imports and variables. PR: <https://github.com/pydantic/pydantic-ai-harness/pull/503>

Mellea | Merged PR #1471*Python, OpenTelemetry tracing, async tests, pytest | Jul 2026*

- Added deterministic mocked-backend tests for async span timing, context propagation, token usage, span lifetime, and consecutive generations. PR: <https://github.com/generative-computing/mellea/pull/1471>

FastStream | Merged PR #2961*Python, FastAPI compatibility, dependency injection, pytest | Jul 2026*

- Restored FastAPI 0.140 compatibility for slotted Dependant objects while retaining native dependency fields and FastStream schema metadata. PR: <https://github.com/ag2ai/faststream/pull/2961>

Additional verified upstream merges | 4 of 8 merged PRs*Correctness, dependency errors, schema generation, workflow scoping | Jul 2026*

- Apache Arrow Rust: <https://github.com/apache/arrow-rs/pull/10486> | Altair: <https://github.com/vega/altair/pull/4089> | FastStream FastAPI: https://github.com/faststream-community/faststream_fastapi/pull/2 | Calkit: <https://github.com/calkit/calkit/pull/1028>

EXPERIENCE**AI Engineer | Nov 2025 - Present***VolyxAI | Independent AI product development*

- Build applied AI systems integrating retrieval, multimodal inputs, voice workflows, model APIs, and structured outputs.
- Develop Python services and n8n workflows with webhooks, validation, retries, logging, human approval, and provider fallback.

Key tools: Python, FastAPI, n8n, REST APIs, webhooks, Azure AI Foundry, OpenAI-compatible APIs, PostgreSQL, Redis**Python Automation Developer | Jan 2023 - Present***Freelance | Independent paid client work*

- Developed streaming utilities for large text datasets, including domain filtering, deduplication, splitting, and list curation.
- Built Telegram integrations and asynchronous services using PostgreSQL, Redis, and Docker; integrated external APIs with explicit error handling, rate limits, and operational logging.
- Developed licensing and key-management components for proprietary Python tools, and published reusable command-line utilities for splitting and sorting large text datasets.

Key tools: Python, scripting, asyncio, Telegram Bot API, aiohttp, PostgreSQL, Redis, Docker, REST APIs**CORE SKILLS**

- Languages: Python, SQL
- AI Engineering: retrieval, multimodal inference, voice AI, structured outputs, provider routing, model evaluation
- Models & Providers: Azure AI Foundry, OpenAI-compatible APIs, Deepgram
- ML & Data: scikit-learn, XGBoost, pandas, feature engineering, temporal evaluation
- Backend & Automation: FastAPI, asyncio, REST APIs, n8n, webhooks, SQLAlchemy
- Data & Infrastructure: PostgreSQL, SQLite, Redis, Docker, GitHub Actions, Linux

EDUCATION

MSc Artificial Intelligence · Northumbria University · September 2026 intake

Bachelor of Technology (BTech), Mathematics · Federal University of Technology, Owerri (FUTO) · 2016–2021

CERTIFICATIONS & TRAINING

- Scientific Computing with Python Developer Certification | freeCodeCamp | May 2026 | [Verify credential](#)
- Google AI Professional Certificate | Google/Coursera | February 2026 | [Verify credential](#)
- Model Context Protocol: Advanced Topics | Anthropic training | March 2026 | [Verify credential](#)